UNIVERSITY OF THE PHILIPPINES CEBU • SPECIAL PROJECT

Non-Overlapping Ensemble Weighting via Möbius Decomposition

A Coopetitive Framework for Diagnosing and Leveraging Model Interactions

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INTRODUCTION

The weighting problem in ensembles

Ensembles combine many models — but how much weight should each one get? The Shapley value is the principled answer, yet it has a structural blind spot.

1

Entanglement gap

The Shapley value mixes a model's standalone quality with its interaction effects into one score — you can't tell why a model is weighted as it is.

2

Double-counting gap

"Fixing" this by adding the Shapley Interaction Index counts the same interaction twice — the SII terms re-add information already inside the Shapley value.

3

Interpretability gap

No standard method outputs a pairwise map of which models are complementary, redundant, or independent within the pool.

No existing ensemble method addresses all three.

Our solution: the Möbius decomposition

The Möbius transform (Harsanyi dividends) splits the ensemble's value into mathematically non-overlapping pieces — by construction, with no double-counting.

$m(\{i\})$

Singleton dividend

Each model's pure standalone value

$m(\{i, j\})$

Pairwise dividend

Surplus (+, complementary) or deficit (–, redundant) from interaction only

→

Coopetitive score

$$s_i = m(\{i\}) + \lambda \cdot \tfrac{1}{2} \Sigma\, m(\{i,j\})$$

λ tunes individual quality vs. interaction structure; softmax → weights

WHAT THIS BUYS US

Interpretability

A pairwise dividend matrix — a readable map of redundancy & complementarity

No double-counting

Individual and interaction terms never overlap

Tunable, simple

One parameter λ; ~15 lines of NumPy, no meta-learner to fit

A four-phase validation design

A comparative experiment: the weighting strategy is the independent variable; AUC-ROC (and Brier score) the outcome. Each phase gates the next.

1

Construct validity

Do pairwise dividends really measure redundancy?

2

Ablation study

Does each component pull its weight?

3

Benchmark comparison

Competitive with established methods?

4

λ sensitivity & analysis

How does λ shape performance?

Pipeline: Train base models → evaluate all 2^n coalitions on validation → compute Möbius dividends → derive coopetitive weights → test on held-out set.

Statistics follow Demšar (2006): Friedman omnibus + Holm-corrected Wilcoxon.

METHODS

Experimental setup

FIVE HETEROGENEOUS BASE MODELS

LR	Logistic Regression	*linear*
RF	Random Forest	*bagging*
SVM	Support Vector Machine	*kernel*
XGB	XGBoost	*boosting*
KNN	k-Nearest Neighbors	*instance*

Five distinct inductive biases — chosen so pairwise dividends have real structure to detect.

THE EVALUATION

15 **binary-classification datasets**

OpenML / UCI — medical, financial, signal, etc. (208 to 48,842 samples)

32 **coalitions per run**

all 2^5 subsets evaluated exactly by AUC-ROC on the validation set

75 **runs per method**

15 datasets × 5 random seeds; 60 / 20 / 20 train-val-test split

The coopetitive weighting formula

$$S_i = m(\{i\}) + \lambda \cdot \tfrac{1}{2} \sum_{j \neq i} m(\{i,j\})$$

$m(\{i\})$

INDIVIDUAL

Standalone contribution of model i.

$\tfrac{1}{2} \Sigma\, m(\{i,j\})$

INTERACTION

Net complementarity / redundancy with the rest of the pool. The ½ shares each pair fairly (Shapley rule).

λ

THE DIAL

$\lambda = 0$ → pure performance-weighting. Larger λ → weights track interaction structure more strongly.

Scores → weights via softmax with adaptive temperature. **Key design choice:** individual-only ($\lambda = 0$) is algebraically identical to performance-weighting — so the whole case for the interaction term rests on whether $\lambda > 0$ beats it (Hypothesis H1).

Four experiments, four hypotheses

H1	**Adding interaction info ($\lambda > 0$) beats performance-weighting ($\lambda = 0$)**	*the pivotal test*
H2	**Pairwise dividends correctly detect known redundancy & complementarity**	*construct validity*
H3	**The framework is competitive with established ensemble methods**	*benchmark ranking*
H4	**A single fixed $\lambda = 0.5$ works well across datasets**	*robustness check*

H1 is the make-or-break question; H2 gates everything by checking the diagnostic actually measures what it claims.

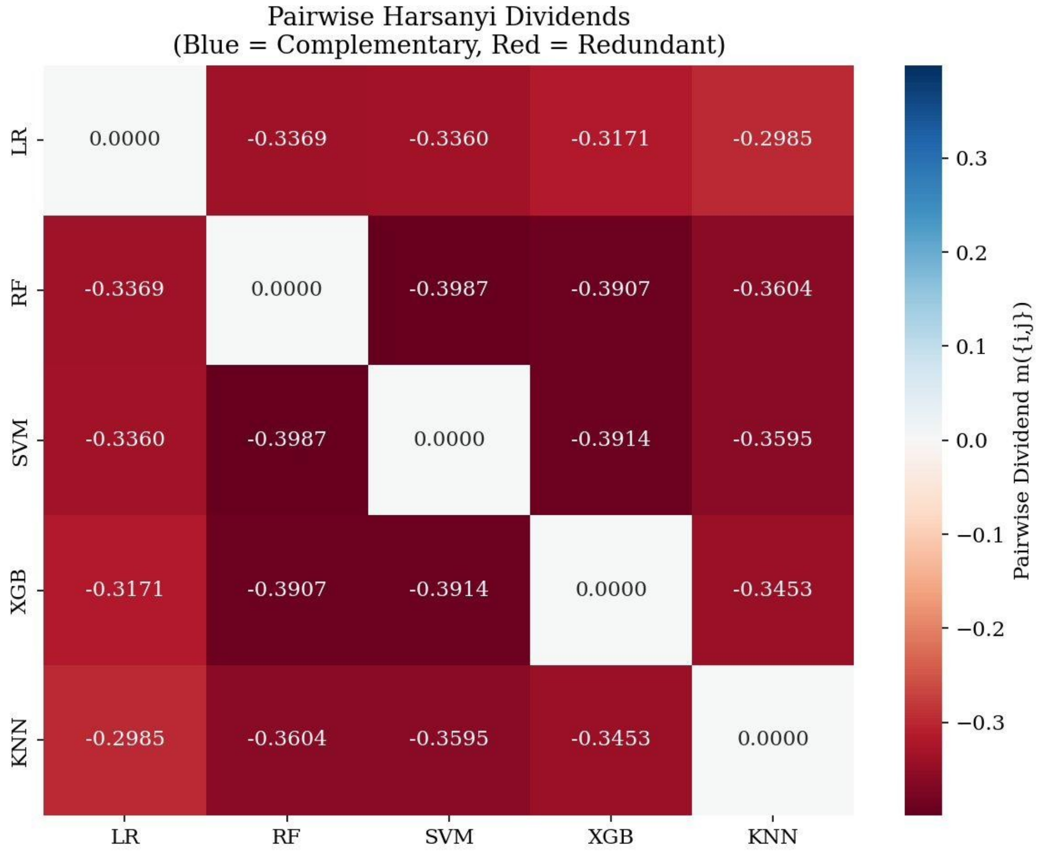
Construct validity: the diagnostic works

FOUR CONTROLLED TESTS

	Test	Result
A	Detects redundancy	PASS[1]
B	Agrees with SII (r = 0.96)	PASS
C	Smooth, monotonic (r = −0.99)	PASS
D	Negative control	FAILED

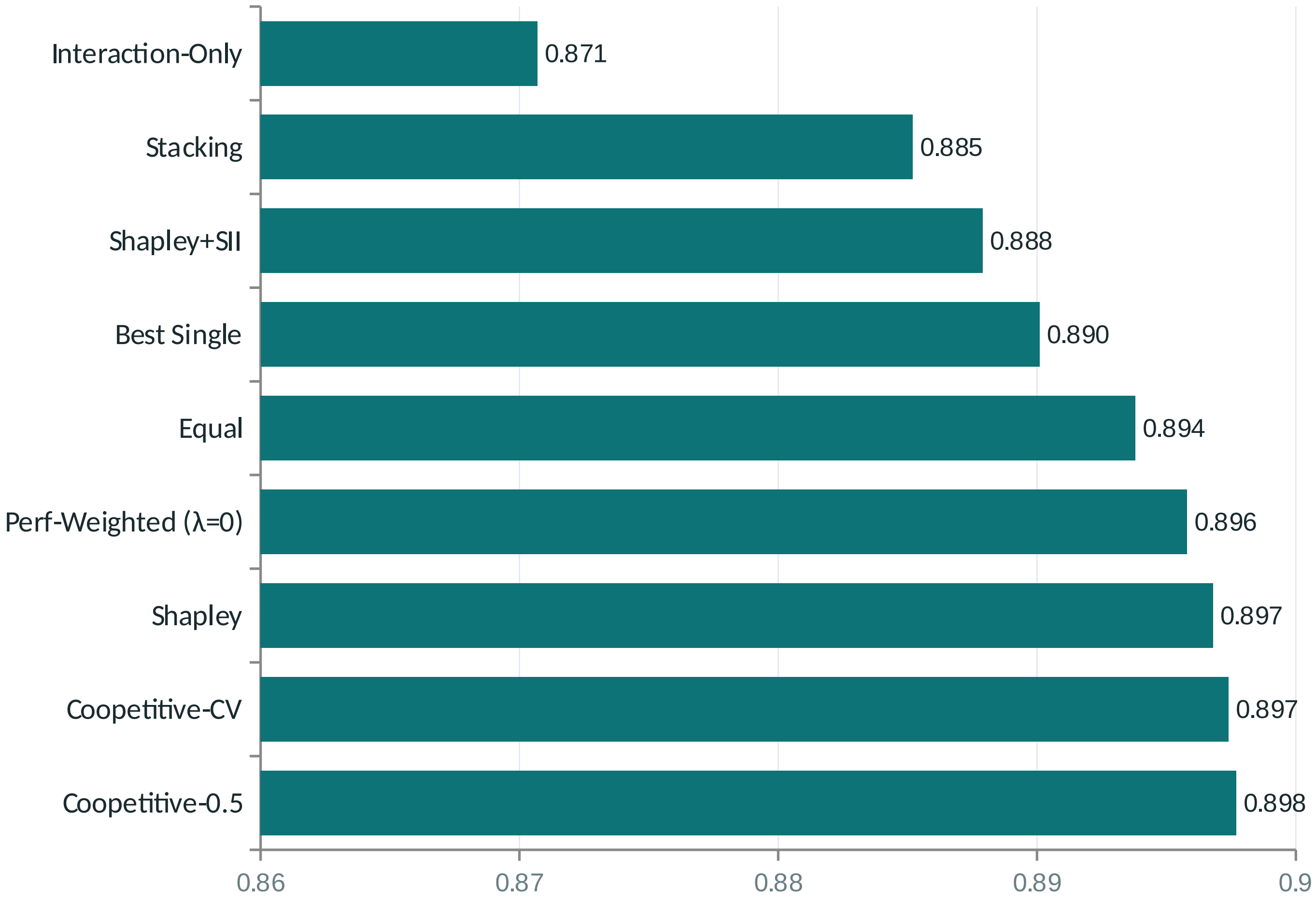
[1] Test A criterion revised post-hoc (provisional). Test D fails criterion 1: max |m| = 0.039 is ~4× the 0.01 threshold — but the mean (0.017) is >5× below Test A, so the excess is finite-sample AUC noise (~160 val. samples), not real interaction.

H2 partially supported — 3 of 4 tests pass convincingly.



Every pair is redundant (negative): most redundant SVM–XGB, least KNN–LR. The matrix is itself the durable contribution.

Does the interaction term help?



Mean AUC-ROC across 15 datasets × 5 seeds (axis zoomed; total spread only 0.027)

Coopetitive-0.5 leads

Best AUC (0.8977) and best calibration (Brier −0.097) of any variant; wins 9 / 15 datasets.

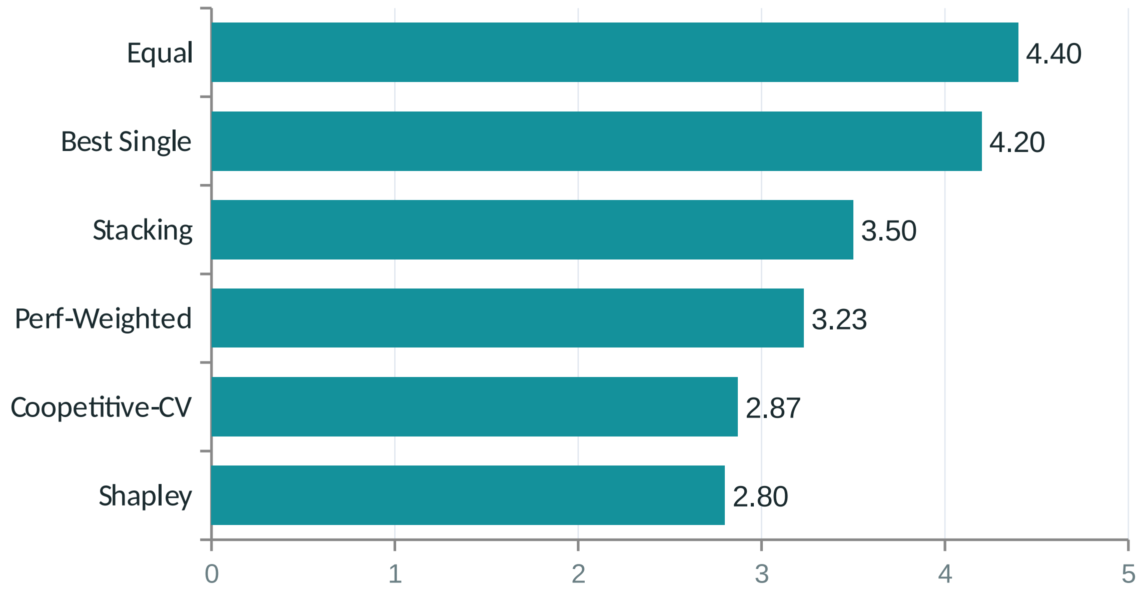
Double-counting, confirmed

Shapley + SII (0.8879) loses 0.009 AUC to plain Shapley — ~⅓ of the whole spread. Re-adding interactions hurts. Möbius avoids this by construction.

H1: not significant. $p = 0.148$, +0.0018 AUC, $r = 0.29$. Directionally consistent but underpowered — a real but small effect.

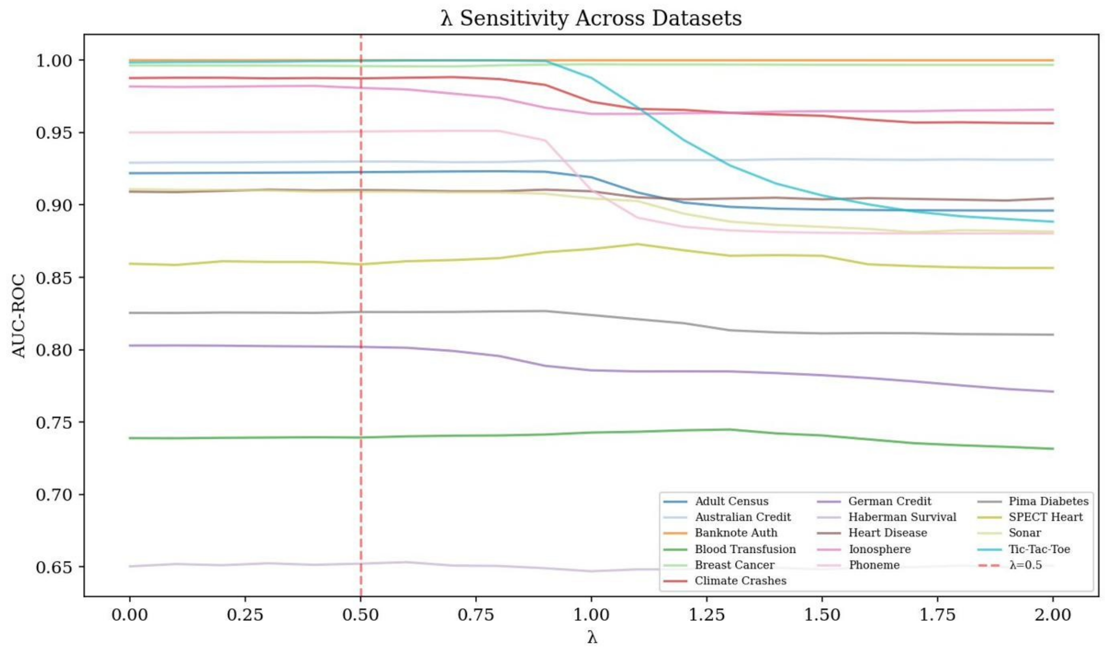
Benchmark ranking & λ sensitivity

BENCHMARK — AVG. RANK (LOWER IS BETTER)



Coopetitive-CV ranks 2nd of 6
— essentially tied with Shapley at the top, ahead of performance-weighting, stacking, and the rest. *H3: not significant (Friedman p = 0.072), but competitive.*

λ SENSITIVITY — NO UNIVERSAL DEFAULT



Optimal λ ranges 0.0–1.5 (mean 0.67). Fixed λ = 0.5 reaches 95% of optimum on only 13% of datasets. *H4 not supported — use cross-validated λ.*

What the evidence says

Hypothesis	Verdict	Key evidence
H1 Interaction term improves performance	Not sig.	$p = 0.148$; +0.0018 AUC; wins 9/15; best Brier
H2 Dividends detect model relationships	Partial	3/4 tests pass; $r = 0.96$ with SII
H3 Competitive with standard methods	Not sig.	Ranks 2nd of 6; Friedman $p = 0.072$
H4 A fixed λ works everywhere	No	Optimal λ spans 0.0–1.5; use CV

THE HONEST READING

Interpretability is the real win

The dividend matrix gives a diagnostic no other method offers — independent of any accuracy gain.

Performance is competitive, not breakthrough

Effects are small; ~30–40 datasets would be needed to settle H1.

Coopetition ran one-sided

All 75 runs gave negative dividends — it acted purely as a redundancy penalty here.

Conclusions & limitations

CONCLUSIONS

- The Möbius decomposition resolves the interpretability gap — the pairwise dividend matrix maps redundancy and complementarity directly.
- It resolves the double-counting gap empirically: Shapley + SII degrades AUC, confirming the core thesis.
- Entanglement (H1) and benchmark (H3) gains were directional but not significant; calibration was best-in-class.

Bottom line

A principled, interpretable diagnostic instrument for ensemble design — transparency that performance-focused methods don't provide.

LIMITATIONS WE KEEP IN VIEW

Pairwise truncation

Higher-order terms dropped; decomposition coverage stayed negative (−2.6 to −5.0).

No fixed λ

Optimal λ is dataset-dependent (0.0–1.5).

Scalability

Exact coalition enumeration limits pools to $n \le 12$.

Scope

Binary classification only; $n = 5$ weakens the rank-stability check.

FUTURE WORK

Where this goes next

Higher-order dividends

Add 3-way+ terms to restore decomposition coverage — the most direct extension.

Dividend-informed pruning

Use the matrix to drop redundant models before combining (grand coalition lost 92% of runs).

Broaden & scale

Multi-class & regression; sampling-based estimation to break the $n \leq 12$ ceiling.

Thank you

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github.com/maxellmilay/mobius-decomposition-ensemble-weighting